

Miscellaneous Points

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ATPG DRCs (K rules, F rules)

F7 :

Decompressors must be controlled by defined channel inputs. F7 traces all inputs to the decompressor to verify that they are connected to channel input pins.

Similarly we get F10, if it happens on compressor side.

K19 :

This design rule check simulates the EDT decompressor netlist by applying pseudorandom patterns and compares the results with the emulated values

K22 :

This design rule check simulates the EDT compactor fed by pseudorandom patterns, and compares the results to the emulated values.

ROM MBIST

The MBIST which is discussed in the course is RAM MBIST.

ROM MBIST has a slight difference because ROM is read only memory. The value is written onto the ROM only once at time of manufacturing. After that we can't change the contents of ROM.

ROM MBIST :

- Designers gives the value that will be written onto the ROM
- *rom_content_file : filename ;*
A property that specifies a ROM content file from which the MISR signature is calculated. If expected_rom_signature is set to “auto”, and the specified file is readable, the process_patterns_specification command calculates the MISR signature from the file
- During ATE testing, the outputs of ROM will be compressed (using MISR – Multi Input Signature Register) and compared with the signature value.
- Read only algorithm is used for ROMs

```
PatternsSpecification(design_name, design_id, pattern_id) {  
  Patterns(pattern_name) {  
    TestStep(name) {  
      MemoryBist {  
        Controller(instance) {  
          MemoryInterface(memory_id) { //repeatable  
            expected_rom_signature : binary | auto ;  
            rom_content_file       : filename ;  
            comparator_id_select   : all | none | integer ;  
          }  
        }  
      }  
    }  
  }  
}
```

Level 3 projects ATPG TDFT script file explanation

ATPG TDF :

Command : < line 21 >

Transition patterns requires the user to define the no. of ATPG cycles in the capture phase. This is because ATPG tool generates patterns based on pulses and doesn't know the freq of fastclk.

-Plt.cycles switch replaces the ATPG cycles with a set no. of external capture cycles using the specified timeframe.